

# Pre-Algebra

## Trimester 2 Study Guide

Name \_\_\_\_\_

Period \_\_\_\_\_

Find each product:

1.  $\frac{5}{9} \times (-\frac{3}{8})$

$$\boxed{-\frac{5}{24}}$$

2.  $(2\frac{1}{4}) \times (-5)$

$$\frac{9}{4} \times -\frac{5}{1} = \boxed{-\frac{45}{4}}$$

Find each quotient:

3.  $\frac{7}{11} \div -\frac{3}{7}$

$$\frac{7}{11} \times -\frac{7}{3} = \boxed{-\frac{49}{33}}$$

4.  $-1\frac{2}{9} \div \frac{4}{5}$

$$-\frac{11}{9} \times \frac{5}{4} = \boxed{-\frac{55}{36}}$$

Solve each equation:

5.  $x - \frac{2}{3} = \frac{4}{5}$

$$x - \frac{10}{15} = \frac{12}{15}$$

$$+ \frac{10}{15} \quad + \frac{10}{15}$$

$$\boxed{x = \frac{22}{15}}$$

6.  $\frac{7}{11}x = \frac{3}{8}$

$$\frac{11}{7} \cdot \frac{7}{11}x = \frac{3}{8} \cdot \frac{11}{7}$$

$$\boxed{x = \frac{33}{56}}$$

7. In Mr. Z's 3<sup>rd</sup> period class there are 15 students from Seminole County, 3 students from Volusia County, and 5 students from Orange County. What is the ratio of students from Orange County to total students in Mr. Z's 3<sup>rd</sup> period class?

$$\frac{5}{15} = \frac{1}{3}$$

8. What is the unit price if a package of 8 candy bars costs \$6.48

$$8 \overline{) 6.48}$$

$$\begin{array}{r} 0.81 \\ 8 \overline{) 6.48} \\ \underline{6.4} \phantom{0} \\ 8 \phantom{0} \end{array}$$

\$.81 per candy bar

9. Solve the proportion:  $\frac{4}{11} = \frac{5}{n}$

$$\frac{4n}{4} = \frac{55}{4}$$

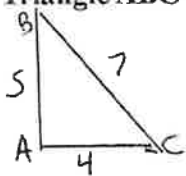
$$\boxed{n = \frac{55}{4}}$$

10. If 1 inch on a map represents 6 miles on the road, how many miles would be represented by 3.5 inches?

$$\frac{6}{1} = \frac{m}{3.5}$$

$$m = 210 \text{ miles}$$

11. Triangle ABC and Triangle DEF are similar. Find the length of x.

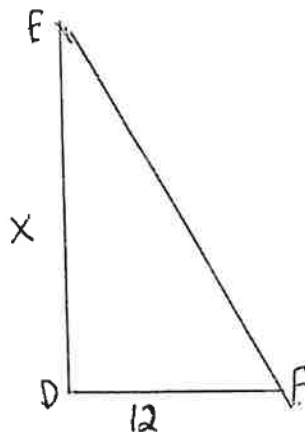


$$\frac{4}{12} = \frac{5}{x}$$

$$\frac{4x}{4} = \frac{60}{4}$$

$$x = 15$$

$$x = 15$$



12. Convert 45% into a decimal and a fraction in lowest terms.

$$.45$$

$$\frac{45}{100} = \frac{9}{20}$$

13. 5 is what percent of 50?

$$\frac{5}{50} = \frac{n}{100}$$

$$n = 10$$

$$10\%$$

14. What is 85% of 140?

$$\frac{n}{140} = \frac{85}{100}$$

$$n = 119$$

15. What is the percentage of change if something dropped from 20 down to 12?

$$\frac{8}{20} = \frac{n}{100}$$

$$n = 40$$

$$40\%$$

16. What is the final cost of an Taylor Swift shirt that was originally priced at \$45.00 on sale 40% off with a 7% sales tax.

$$45 \times .4 = 18 \quad \text{OR} \quad \frac{n}{45} = \frac{40}{100}$$

$$n = 18$$

$$45 - 18 = 27.00$$

$$27 \times .07 = 1.89 \quad \text{OR} \quad \frac{n}{27} = \frac{7}{100}$$

$$n = 1.89$$

$$27.00 + 1.89 = 28.89$$

Find the measures of center of the following data set:

9 11 10 13 8 4 8 10 8  
4 8 8 8 9 10 10 11 13

17. Mean  $\frac{81}{9} = 9$

18. Median  $= 9$

19. Mode  $= 8$

20. Range  $13 - 4 = 9$